

Advanced Web Technologies - Individual Project Assignment

Online Bookshop System with API Integration

1. Assignment Overview

Design and develop a JSON-based web application for an online bookshop that integrates with external APIs and implements user authentication.

2. Learning Objectives

- Understand and implement RESTful API concepts
- Work with JSON data structures
- Implement user authentication and authorization
- Develop frontend and backend integration
- Handle API calls and responses
- Manage user sessions and data persistence

3. Project Requirements

3.1 Technical Requirements

- Frontend Development:
 - HTML5, CSS3, and JavaScript
 - Any modern frontend framework (React.js, Vue.js, or Angular)
 - Responsive design implementation
 - Form validation
- Backend Development:
 - RESTful API implementation
 - JSON data format for all API responses
 - Database integration (MongoDB/MySQL)
 - User session management

3.2 Functional Requirements

1. User Authentication Module

- User Registration (Sign Up):
 - Username
 - Email address
 - Password (with encryption)

- Basic profile information
- Form validation
- Email verification (optional)

2. User Login:

- Email/Username authentication
- Password verification
- Session management
- Remember me functionality

Book Management Module

1. Book Listing:

- Integration with external book API
- Display book details (title, author, price, etc.)
- Implement search functionality
- Filter and sort options

2. Book Details:

- Detailed view of each book
- Book description
- Price information
- Stock availability

Shopping Module

1. Shopping Cart:

- Add/remove books
- Update quantities
- Calculate total price
- Cart persistence

2. Checkout Process:

- Shipping information
- Payment method selection
- Order confirmation
- Invoice generation

4. API Integration Requirements

1. External API Integration:

- Implement connection with book data API (e.g., Google Books API, Open Library API)
- Handle API authentication
- Implement error handling
- Cache API responses

2. Internal API Development:

- Create RESTful endpoints for:
- User management
- Shopping cart operations
- Order processing
- Book management

5. Security Requirements

- Implement JWT (JSON Web Tokens) for authentication
- Password encryption
- Input validation and sanitization
- Protection against common web vulnerabilities
- Secure session management

6. Deliverables

1. Source Code:

- Well-documented code
- README file with setup instructions
- Requirements.txt or package.json

2. Documentation:

- System architecture diagram
- API documentation
- Database schema
- Setup and deployment guide

3. Presentation:

- Live demonstration
- Code walkthrough
- Technical decisions explanation

7. Evaluation Criteria

- **Code Quality (25%)**
 - Clean, well-organized code
 - Proper error handling
 - Code comments and documentation
- **Functionality (30%)**
 - All required features working
 - API integration success
 - User experience
 - Project Demonstration with Functionality presentation
- **Security Implementation (20%)**
 - Authentication/Authorization
 - Data protection
 - Security best practices
- **Technical Documentation (15%)**
 - Clarity and completeness
 - API documentation
 - Setup instructions
- **Innovation and Creativity (10%)**
 - Additional features
 - UI/UX design
 - Performance optimization

8. Submission Guidelines

Include all documentation and code folder in the one folder as ZIP file

Submit all code with finalized document to LMS

live demo on Faculty scheduled date.

9. Timeline

Project Duration: 2 weeks (14 Days)

14 Days milestones:

- Setup and basic structure
- API integration and user authentication
- Shopping features and testing
- Documentation and final touches

10. Additional Notes

- Students are encouraged to use modern development tools and practices
- Regular commits to version control are expected
- Code plagiarism will result in automatic failure
- Questions and clarifications can be addressed during lab sessions

This assignment will help students understand the practical implementation of web technologies while creating a real-world application. The focus is on both technical implementation and professional development practices.

Documentation Guidelines

Documentation Requirements for Online Bookshop Assignment

Students must submit detailed documentation including the following components:

1. User Interface Documentation Requirements

1.1 User Authentication Screens

Students must document:

- Registration page screenshot
- Login page screenshot
- Associated HTML/CSS/JavaScript code screenshots for each page
- Authentication success/error messages screenshots

1.2 Book Listing and Search

Required documentation:

- Main book listing page screenshot
- Search functionality interface screenshot
- Filter/Sort options screenshot

- Book details view screenshot
- Code screenshots for API integration
- Search implementation code screenshots

1.3 Shopping Cart

Document the following:

- Cart view screenshot
- Add to cart functionality screenshot
- Update quantity interface screenshot
- Remove items interface screenshot
- Cart total calculation screenshot
- Related code screenshots for cart management

1.4 Checkout Process

Include:

- Checkout form screenshot
- Payment interface screenshot
- Order confirmation page screenshot
- Implementation code screenshots
- Order processing code screenshots

1.5 User Profile

Document:

- Profile view screenshot
- Edit profile interface screenshot
- Order history view screenshot
- Related implementation code screenshots

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